

Yoga as an Eco-therapy for Achieving United Nations Sustainable Development Goals: An Explorative Study

Abstract

Context: The SDGs highlight the connections between the environmental and socioeconomic aspects of sustainable development. *Yoga* not only results in one's physical and mental resilience but also develops positive feelings toward one's environment to help protect it from destruction. **Aims:** The study aimed to understand the role of *Yoga* for achieving the United Nations SDGs and to draft a policy statement for its implementation to achieve the SDGs. **Methodology:** The study was conducted in three phases. Phase 1 included a review of available literature. Phase 2 included contacting experts from *Yoga*, community health, social work, and medicine from all over India and across the globe for discussion. Phase 3 included drafting the recommendations to be submitted to the CCRYN based on the results from the first two phases. **Results:** A total of 320 research articles (217 included) were screened. A total of 103 were excluded for reasons (being not relevant, inability to get full text of articles, and outcome of the study not matching). Phase 2 was conducted in three sessions with different groups of experts for their input. In phase 3, recommendations were finalized, and the policy statement was drafted. **Conclusions:** *Yoga* can help in achieving the targets of the UN SDGs. The recommendations, if implemented, can solve the issues endangering nature and the ecosystem. Thus, *Yoga* builds a harmony between humans and nature.

Keywords: *Eco-therapy, United Nations Sustainable Development Goals (UN SDGs), Yoga*

Introduction

The United Nations (UN) 17 Sustainable Development Goals (SDGs) aim toward promoting community and environmental sustainability.^[1] The environment of a human being often can have an impact on their mental health and well-being. Scientific studies have well established that exposure to nature is positively associated with numerous aspects of both physiological and psychological health, including positive impacts on anxiety, ADHD, aggression, depression, PTSD, and stress.^[2-4] This synergistic relationship between a human being and their natural environment is an important aspect of environmental sustainability. Thus, methods of promoting natural lifestyles that are conducive to environmental sustainability need to be explored.^[5]

Yoga teaches one to lead a simple, minimalistic life with contentment that helps reduce dependency on materialistic pleasures, comforts, carbon footprint, and

pollution. It also has a direct bearing on one's health. *Yoga* has a strong connection with nature and requires no artificial facilities for practice. It is already widely practiced and has a dedicated following worldwide, especially after the declaration of the International Day of *Yoga* on June 21.^[6] *Yoga* encompassing the eight limbs of *Yoga* such as *yama* (abstinences), *niyama* (observances), *asana* (postures), *pranayama* (breathing), *pratyahara* (withdrawal), *dharana* (concentration), *dhyana* (meditation), and *samadhi* (absorption),^[7] if followed as a lifestyle, can be an effective intervention in dealing with multiple SDGs and in promoting community and environment sustainability.

Outdoor activities in nature help in improving and developing resilience in one's own mental and physical health.^[8] In the same context, *Yoga* (connecting with nature), even in unfavorable environmental conditions, is beneficial.^[9] *Yoga* through successive stimulation-relaxation helps

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break the loop of uncontrolled speed of thoughts (stress), helps gain control over the mind, and harmonizes the disturbances at each of the five levels (physical, energy/vital, mental and emotional, intellectual, blissful, and spiritual sheaths) to tackle psychosomatic problems.^[10] *Yoga* can be used as a powerful method to not only develop one's physical and mental resilience to deal with environmental calamities but also to develop positive feelings toward one's environment, to help protect it from destruction.^[11] Thus, if *Yoga* is advocated to all individuals through inclusive approaches of community participation, it could be an aid in the promotion of environmental and community sustainability (SDG Goal 11), which would help achieve many SDGs. *Yoga* would help transform the community into ambassadors of change enthused about the cause of ecology through simple, minimalistic living. In this context, this study aimed to understand how *Yoga* can be used to achieve the United Nations' SDGs and to draft a policy statement to implement *Yoga* to achieve them.

Methodology

It was an exploratory study design and, therefore, did not include any sample size, intervention, and data analysis. The study was conducted in three phases: Phase 1 included review of literature, Phase 2 included brainstorming with experts, and Phase 3 included recommendations, policy drafting, and report submission to CCRYN.

Phase 1: Review of literature

The process of review of literature in Phase 1, included a search of "*Yoga*" related studies in each SDG (The keywords were prepared for each sub-target of all the seventeen SDGs using *Yoga* as an intervention, for example, *Yoga* for eradicating poverty/zero hunger/gender equality/climate action/education, etc.) using databases PubMed, Scopus, Cochrane Library, and Google Scholar between July 2023 and September 2023. Articles were screened after going through the title, abstract, and the complete paper. Relevant data from the articles were saved in the Microsoft Excel sheet (Microsoft corporation, Redmond, Washington, United States of America).

Phase 2: Brainstorming with the experts

Three brainstorming sessions were conducted with the experts, (a) In-Person stakeholder's workshop, (b) Presentation and discussion with the members of the Indian *Yoga* Association, and (c) Online stakeholder's discussion across different countries.

In-person stakeholder's workshop

A brainstorming discussion was done with twenty experts from different schools of *Yoga*, such as (*Patanjali Yoga* University, Haridwar; *Krishnamacharya Yoga Mandir*, Chennai; *Kaivalyadhama*, Pune; *Iyengar School of Yoga*, Mumbai; and *SVYASA Yoga* University, Bengaluru). Apart from *Yoga* experts, other experts involved were from different fields

such as medicine, social work, environment, naturopathy, and *ayurveda*. It helped us to understand how *Yoga* can be used to achieve the UN SDGs and to prepare a white paper.

Presentation of "Yoga for achieving UN SDGs" at National Conclave organized by Indian Yoga Association in collaboration with Art of Living Headquarters, Bengaluru

An in-person presentation was done on the draft of recommendations generated from phase 1 in front of the members of the Indian *Yoga* Association (IYA) during a National Conclave at Art of Living Ashram, Bengaluru, and discussed how IYA can participate to achieve them. The number of delegates, apart from the members of IYA who attended the conclave, was more than 100 from various backgrounds such as *Yoga*, social work, educationists, environmentalists, and researchers.

Online stakeholders discussion

An online stakeholders' discussion was conducted on *Yoga* for achieving UN SDGs with ten experts from *Yoga* schools and associations from different countries. Experts were from The United States of America, Bahrain, Portugal, Italy, Japan, South Korea, Singapore, Germany, Brazil, and Vietnam. The experts were presented with the draft recommendations and were asked for their inputs/suggestions/modifications, if any, for a particular SDG.

Phase 3: Policy drafting (recommendations)

A draft on recommendations was prepared and circulated among all the experts involved in phase 2 to ask for their inputs/comments/consent for finalizing the recommendations before submission to the Central Council for Research in *Yoga* and Naturopathy (CCRYN) under the Ministry of AYUSH, Government of India.

Results

Phase 1: Review of literature

The articles were screened by two separate experts (AJ and VK), and the final decision about the inclusion/exclusion of the articles was made. Reasons for excluding the articles were: Does not directly talk about the related topic, inability to get full text of articles, outcome of the study not matching, not related to *Yoga*, etc. None of the articles referred to more than one SDG. The results of the search are given in Table 1.

Overall, the review revealed the role of *Yoga* in various SDGs as explained further. For SDG 3, we got the maximum relevant studies. *Yoga* has a strong potential as an intervention for the maintenance of health and curing ailments. *Yoga* is used in education for ethical learning in SDG 4 and SDG 13; *Yoga* as a source of earning in SDG 1, SDG 2, and SDG 8; *Yoga* as a cultural contribution in SDG 5 and SDG 8; *Yoga* as a tool to spread peace, harmony in nature and health across the community and globe in SDG 13, SDG 15, SDG 16 and SDG 17.

Table 1: Number of articles in their respective sustainable development goals

SDGs	Articles screened	Articles included	Articles excluded
SDG 1	70	40	30
SDG2	23	3	20
SDG 3	107	97	10
SDG 4	8	4	4
SDG 5	10	5	5
SDG 6	12	9	3
SDG 7	0	0	0
SDG 8	15	10	5
SDG 9	8	2	6
SDG 10	0	0	0
SDG 11	2	2	0
SDG 12	7	5	2
SDG 13	16	7	9
SDG 14	3	2	1
SDG 15	11	8	3
SDG 16	21	18	3
SDG 17	7	5	2
Total	320	217	103

SDGs: Sustainable development goals

Phase 2: Brainstorming with the experts

The raw inputs from Phase 1 of the study were refined with the help of collective inputs from three brainstorming sessions with the experts from India and across the globe and were merged together to form the below recommendations in each SDG [Table 2].

Phase 3: Policy drafting (recommendations):

A policy statement was drafted as “Recommendations on *Yoga* for achieving United Nations’ Sustainable Development Goals (UN SDGs) after receiving consent from all the experts involved in Phase 2 of the study. It was submitted to the Central Council for Research in *Yoga* and Naturopathy (CCRYN) under the Ministry of AYUSH, Government of India, so that it can be implemented nationwide (A detailed report of the recommendations is available with the authors on request).

Discussion

Based on the recommendations on how *Yoga* can help achieve the UN SDGs, a few achievable outcomes were suggested for implementation:

Yoga is a way of life: All through the discussion one point that has been prominent is that *Yoga* is a way of life (following *Yoga* philosophy and *Yogic* practices), which can help in achieving the UN SDGs. Hence, beyond *Yoga* practices of *asana*, *pranayama*, and meditation, it is important to look at *Yoga* philosophy and concepts to help achieve the UN SDGs. Several concepts from *Yoga* philosophy can be indirectly applied to all the SDGs, for example, *swadhyaya* (self-reflection),

mindfulness (Awareness of the present moment), *trigunas* (three qualities in every individual), *tapas* (self-discipline). However, in this document, we have only mentioned the *Yoga* philosophy concepts that can directly help achieve the specific UN SDG. The recommendations need to be read and interpreted in the above context. The concepts of *Yoga* go beyond the ideas of the SDGs by the UN, which gets revised every 10 years. *Yogic* way of life remains relevant beyond the targets and time frame of the UN SDGs.

1. *Incorporating recommendations in existing projects:* The recommendations in this document can be incorporated into the existing campaigns and programs of the UN. For example, several UN agencies have projects on reducing food wastage with circular economy principles, projects for women, resilience of vulnerable individuals, etc., where these recommendations on *Yoga* and *Yoga* practices to achieve the specific SDGs can be incorporated. These recommendations can be linked to tangible outcomes that can help countries achieve the SDGs through *Yoga*. The current team would be happy to volunteer to review all UN projects related to each of the SDGs and provide specific guidance on how these recommendations can be woven into the existing projects.
2. *Collaborative research for implementation of recommendations:* Efficacy of *Yoga* has already been validated in several thrust areas of the UN, such as prevention of violence, public health systems, etc. Over a period, some of these SDG-specific recommendations can be developed into new projects/programs for testing and implementation across countries.
3. *Convene discussions on Yoga for SDGs:* The UN can convene discussions and round tables on *Yoga* for achieving SDGs as a lead-up to a Joint UN conference on *Yoga* for achieving SDGs for International *Yoga* Day 2025. These discussions would help finalize the way forward in research and implementation of the recommendations across countries.

Conclusion

Yoga has an important role in achieving the UN SDGs. The recommendations, if implemented nationwide, help solve the issues endangering nature and the ecosystem, thus building a harmony between humans and nature.

Ethical statement

The study was approved by Institutional ethics committee at NIMHANS.

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Table 2: Relevant yogic concepts and recommendations in each United Nations sustainable development goals

SDG number	Name	Yogic concepts	Recommendations
SDG 1	No poverty	<i>Dana</i> <i>Aparigraha</i> <i>Santosha</i>	Vocational opportunities for income generation Charity oriented Lifestyle Self-sustaining <i>yoga</i> villages
SDG 2	Zero hunger	<i>Mitahara</i> <i>Dana</i> <i>Asteya/Aparigraha</i>	Developing cultural and eco-friendly food habits <i>Mitahara</i> and promoting food donations
SDG 3	Good health and well being	<i>Astanga/Hatha Yoga</i> <i>Jnana Yoga</i> <i>Karma Yoga</i> <i>Raja Yoga</i> <i>Bhakti Yoga</i> <i>Panchakosha viveka</i>	Specialized <i>yoga</i> programs for different health disorders <i>Yoga</i> for well-being of vulnerable groups (women and youth) <i>Yogic</i> counseling Accessibility and inclusiveness of <i>Yoga</i>
SDG 4	Quality education	<i>Yama-Niyama</i> <i>Brahmacharya</i> <i>Sravana, Manana, Nidhidhyasana</i> <i>Yoga Anushasana</i>	<i>Yoga</i> as a part of the education curriculum <i>Yoga</i> for value education <i>Yoga</i> as extracurricular activities <i>Yoga</i> research
SDG 5	Gender equality	<i>Triguna</i> <i>Vasudaiva Kutumbakam</i>	<i>Yoga</i> for All Inculcating <i>Yogic</i> qualities for gender equality
SDG 6	Clean water and sanitation	<i>Saucha</i> <i>Tapas</i> <i>Aparigraha/Asteya</i> <i>Swadhyaya/Mindfulness</i>	Awareness programs Community water-based projects Research initiatives
SDG 7	Affordable and clean energy	<i>Vasudaiva Kutumbakam</i> <i>Tapas</i> <i>Aparigraha/Asteya</i> <i>Swadhyaya</i>	<i>Yoga</i> -related activities Mindful consumption and energy awareness Research initiatives
SDG 8	Decent work and economic growth	<i>Karma Yoga</i> <i>Swadhyaya</i> <i>Tapas</i> <i>Asteya/Aparigraha</i>	<i>Yoga</i> as a profession/work <i>Yoga</i> for employers
SDG 9	Industry, innovation, and infrastructure	<i>Triguna</i> <i>Aparigraha</i> <i>Karma Yoga</i>	Innovation in wellness technology Investment in stable infrastructure Increasing the accessibility of small-scale <i>Yoga</i> industries and enterprises in developing countries
SDG 10	Reduced inequalities	<i>Vasudaiva Kutumbakam</i>	<i>Yoga</i> for all Standardized <i>Yoga</i> programs for all age groups Promoting <i>Vasudaiva Kutumbakam</i> Mass <i>Yoga</i> programs
SDG 11	Sustainable cities and communities	<i>Vasudaiva Kutumbakam</i> <i>Santosha</i> <i>Tapas</i> <i>Swadhyaya</i>	Community involvement Awareness programs

Contd...

Table 2: Contd...

SDG number	Name	Yogic concepts	Recommendations
SDG 12	Responsible consumption and production	<i>Ahimsa</i> <i>Saucha</i> <i>Swadhyaya</i> <i>Pratyahara</i> <i>Brahmacharya</i> <i>Aparigraha/Asteya</i> <i>Santosha</i> <i>Yama, ahimsa</i> <i>Niyama- swadhyaya</i>	Mass public awareness Promoting mindful consumption Promoting a sustainable lifestyle
SDG 13	Climate action	<i>Triguna concept</i> <i>Ahimsa</i> <i>Saucha</i> <i>Swadhyaya</i> <i>Pratyahara</i> <i>Santosha</i>	Conducting awareness programs through Government campaigns via social media and in physical events promoting outdoor <i>Yoga</i> practice Promoting <i>Yogic</i> lifestyle and vegetarianism Research initiatives
SDG 14	Life below water	<i>Vasudaiva Kutumbakam</i> <i>Ahimsa</i> <i>Asteya</i>	Awareness and advocacy research initiatives
SDG 15	Life on land	Mindfulness <i>Triguna</i> <i>Vasudaiva Kutumbakam</i> <i>Saucha</i> <i>Ahimsa</i> Mindfulness <i>Asteya</i>	Awareness and advocacy research initiatives
SDG 16	Peace, justice and strong institutions	<i>Ahimsa</i> <i>Anasakti bhava</i> <i>Satya</i> <i>Santosha</i> <i>Vasudaiva Kutumbakam</i>	Conducting workshops on life skills in schools and colleges <i>Yoga</i> workshops Education and training of <i>Yogic</i> concepts
SDG 17	Partnerships for the goals	<i>Vasudaiva Kutumbakam</i>	Promoting cross-sectional collaborations, global networks and alliances Local and global outreach programs Developing value-based relationships in partnerships

SDG: Sustainable development goal

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Conflicts of interest

There are no conflicts of interest.

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